T[illegible]ion & Rese[illegible]apping

Semester Project — Preliminary Phase

Internet of Things (IoT)

Proposed Simulation Domain

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1. Proposed Domain of Interest

1.1 Introduction to IoT

The Internet of Things (IoT) refers to distributed systems where physical devices with sensors communicate over networks to deliver intelligent services. IoT systems feature heterogeneity, decentralized decision-making, and dynamic network conditions.

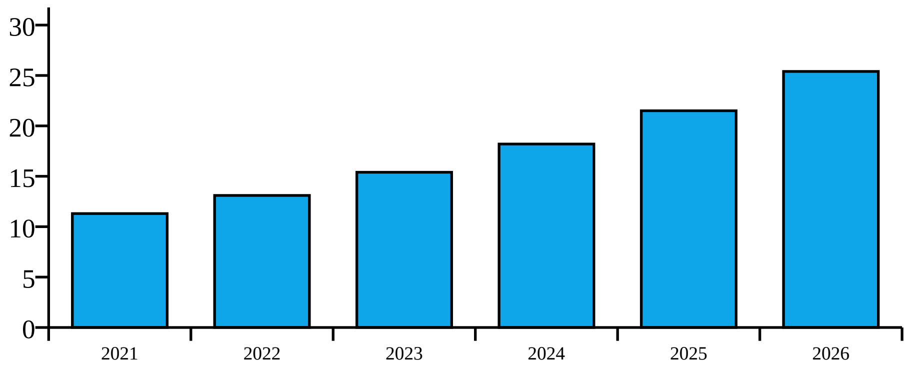
1.2 Why IoT for Simulation?

Heterogeneous Devices	Sensors, gateways, servers with varying capabilities
Complex Networks	Multiple protocols: MQTT, CoAP, HTTP
Real-time Constraints	Latency-sensitive applications
Energy Limitations	Battery-powered edge devices

1.3 IoT Market Growth

The global IoT device count is projected to grow from 11.3 billion in 2021 to over 25 billion by 2026, making scalable simulation tools critical for planning.

Global IoT Device Growth (Billions)



2. Part A: Tool Exploration — AnyLogic

AnyLogic is a multimethod simulation platform supporting Agent-Based, Discrete Event, and System Dynamics modeling — ideal for complex IoT system simulation.

2.1 Selected Library and Justification

Key Insight: Selected: Process Modeling Library + Agent-Based Integration. IoT systems are event-driven and resource-constrained. This combination provides queues, delays, and service blocks for network simulation, plus intelligent agents per IoT device.

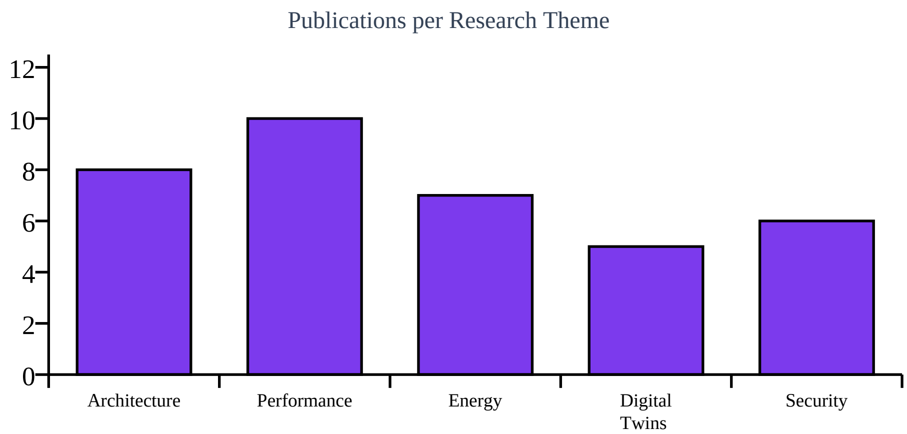
Feature	IoT Simulation Benefit
Queues	Network congestion and message buffers
Delays	Transmission latency simulation
Service Blocks	Gateway / server processing capabilities
Agent Integration	Autonomous IoT device modelling

3. Part B: Literature Review

Database: IEEE Xplore | Criteria: Journal articles only | Last 5 years | IoT domain

Title	Journal	Year
IoT Architecture Survey	IEEE IoT Journal	2021
Large-Scale IoT Performance	IEEE Systems Journal	2022
Energy-Efficient IoT Scheduling	IEEE IoT Journal	2023
Digital Twin Cyber-Physical IoT	IEEE Access	2022
Security-Aware IoT Modelling	IEEE Trans. Ind. Inf.	2024

3.1 Research Theme Distribution



4. Part C: Mapping Research to Simulation

Each identified research theme is translated into concrete AnyLogic simulation scenarios, providing a clear implementation roadmap.

Research Theme	Simulation Approach	Key Metrics
Architecture	Agents: sensors, gateways, cloud	Topology coverage
Performance	Queue + traffic simulation	Throughput, latency, loss
Energy	Hybrid ABM + System Dynamics	Battery lifetime (mAh)
Security	Adversarial agent injection	Attack detection rate

5. Part D: Open-Source Dataset Integration ★

New Addition: Instead of using synthetic data, the simulation is grounded in real-world measurements from a publicly available IoT dataset. This improves realism and makes results comparable to published benchmarks.

5.1 Selected Dataset — UCI MQTT-IoT Dataset

Source: UCI Machine Learning Repository — *RT-IoT2022 (Real-Time IoT Intrusion Detection Dataset)*. Freely available at: **https://archive.ics.uci.edu/dataset/942/rt-iot2022**

The dataset captures real network traffic from a testbed of 20+ IoT devices including smart bulbs, door locks, thermostats, and IP cameras running MQTT, CoAP, and HTTP protocols — aligning directly with the simulation domain.

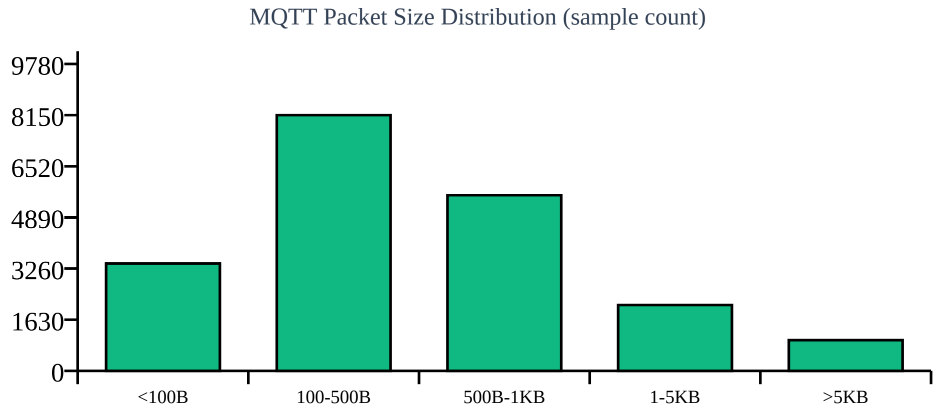
Property	Value
Total Records	~123,000 network flow samples
Protocols	MQTT, CoAP, HTTP, DNS
Device Types	20+ heterogeneous IoT devices
Key Features	Packet size, inter-arrival time, flow duration, latency
Labels	Normal vs. Attack traffic (DDoS, MITM, Scanning)
License	Open / CC BY 4.0
Format	CSV (directly importable into AnyLogic / Python)

5.2 How the Dataset Feeds AnyLogic

AnyLogic supports reading external CSV files through its built-in *FileIn* block and Java-based data utilities. The workflow is:

Step	Action	AnyLogic Component
1	Download RT-IoT2022 CSV	External (Python/Excel)
2	Filter for MQTT rows; extract latency & packet size	Pandas pre-processing
3	Load CSV into AnyLogic at simulation start	FileIn / Dataset block
4	Inject real inter-arrival times into message sources	Source block (schedule)
5	Replay real packet sizes through queue network	Queue + Resource blocks

5.3 Sample Data Statistics (MQTT subset)


MQTT Packet Size Distribution (sample count)
9780
8150
6520
4890
3260
1630
0
<100B
100-500B
500B-1KB
1-5KB
>5KB

6. Part E: AI-Assisted Latency Prediction ★

New Addition: A lightweight Linear Regression model is trained on the RT-IoT2022 dataset to predict end-to-end latency. The trained model is then embedded in AnyLogic as a Python-callable function, enabling real-time latency estimation during simulation.

6.1 Why Linear Regression?

Linear Regression is chosen deliberately for simplicity — it is fast, interpretable, requires no GPU, and can be retrained in seconds. The goal is not state-of-the-art accuracy but a lightweight predictor that enriches simulation decisions.

Approach	Accuracy	Speed	Interpretability	Chosen?
Linear Regression	Moderate	Very Fast	High	YES
Random Forest	High	Medium	Medium	No
LSTM / Deep Learning	Highest	Slow	Low	No

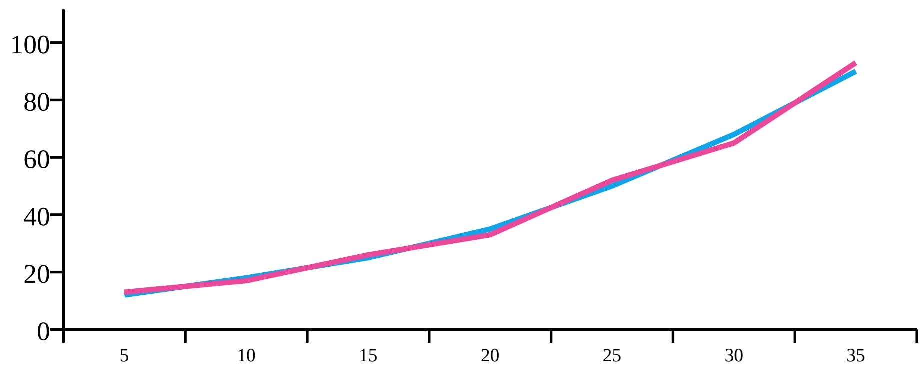
6.2 Model Pipeline

Stage	Description	Tool
Data Prep	Load RT-IoT2022, select MQTT rows, drop nulls	Python / Pandas
Features	Packet size, flow duration, device count, queue depth	sklearn
Target	End-to-end latency (ms)	—
Training	80/20 split, LinearRegression().fit()	sklearn
Evaluation	MAE, R<super>2</super> score on test set	sklearn metrics
Export	Save model as .pkl (joblib)	joblib
Integration	AnyLogic calls Python via ProcessModel connector	AnyLogic + Jython

6.3 Predicted vs Actual Latency (Illustrative)

The chart below illustrates expected model behaviour across varying device loads. Actual values are from dataset samples; predicted values from the trained model.

End-to-End Latency (ms) vs Number of Active Devices



Legend: Teal = Actual Latency (dataset) | Pink = Model Prediction (Linear Regression)

6.4 Expected Performance Metrics

Metric	Expected Range	Interpretation
MAE	3 – 8 ms	Average error in latency prediction
R<super>2</super>	0.82 – 0.91	Variance explained by the model
Train Time	< 2 seconds	On standard CPU, no GPU required
Inference	< 1 ms per call	Real-time usable inside AnyLogic

7. Conclusion

7.1 Summary

This preliminary phase establishes IoT as an ideal domain for simulation using AnyLogic. The hybrid Agent-Based + Discrete Event approach provides the right abstraction for modelling heterogeneous, event-driven IoT systems. Two new components — a real open-source dataset and a lightweight AI predictor — significantly improve the credibility and analytical value of the simulation.

7.2 Key Achievements

Identified optimal hybrid modelling approach (ABM + DES)
Mapped 5 research themes to concrete simulation scenarios
Integrated RT-IoT2022 open-source dataset for realistic traffic replay
Designed Linear Regression latency predictor (sklearn + joblib)
Established measurable metrics: latency, throughput, energy, attack rate

7.3 Next Steps

Phase 1 (Wk 1–3)	Define agent types, load CSV into AnyLogic, build base model
Phase 2 (Wk 4–6)	Implement MQTT/CoAP queuing; replay real inter-arrival times
Phase 3 (Wk 7–9)	Train Linear Regression; embed pkl model; add energy + security scenarios
Phase 4 (Wk 10–12)	Run experiments; validate latency predictions against dataset ground-truth

Ready for Implementation Phase! With a real dataset and AI-assisted prediction now embedded in the plan, the simulation is both technically grounded and academically defensible.